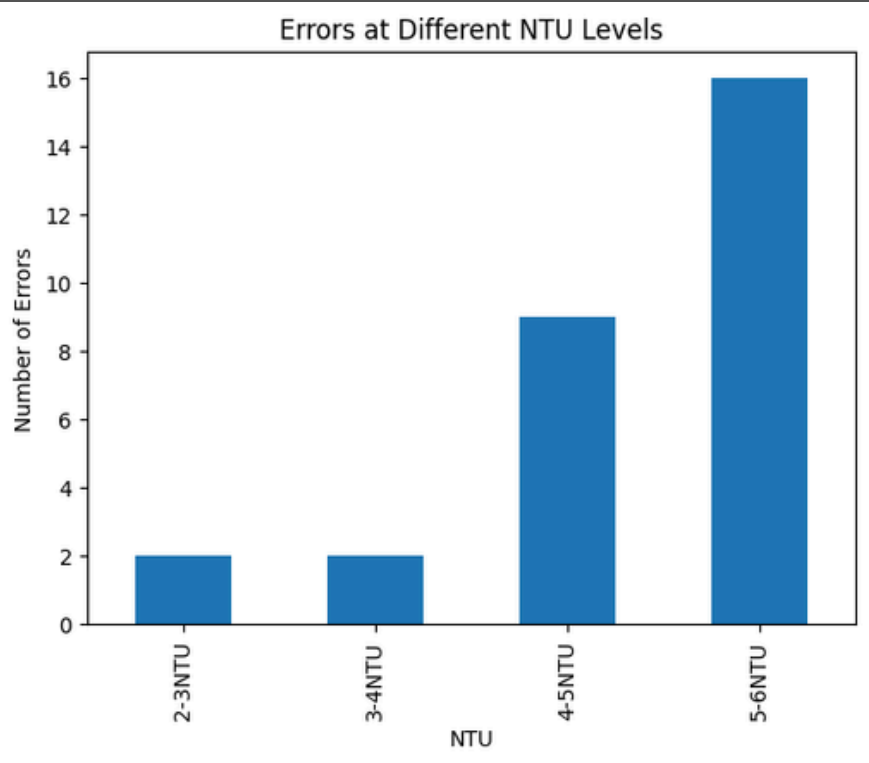


Day 25 - Detailed Error / Failure Analysis

Error Analysis Summary

Metric	Value
Total Test Samples	120
Correct Predictions	91
Misclassified Samples	29
Accuracy	75.83%

NTU-wise Error Distribution



NTU Level	Number of Errors
2-3 NTU	2
3-4 NTU	2
4-5 NTU	9
5-6 NTU	16

Key Findings :

- Most prediction errors occurred at higher turbidity levels (4–6 NTU).
- Error count increased significantly as NTU increased.
- Low turbidity conditions (2–4 NTU) showed very few classification errors.
- High turbidity causes greater optical signal attenuation and scattering, leading to overlap between bit 0 and bit 1 signal values.
- The model performs reliably under low and moderate turbidity but struggles under severe channel degradation.

Failure Patterns Identified :

1. **High NTU Conditions:** Increased scattering and absorption distort the received optical signal.
2. **Signal Overlap:** Analog values of transmitted bit 0 and bit 1 become less separable.
3. **Limited Dataset Size:** Small dataset size restricts model generalization under challenging conditions.

Limitations :

- Dataset contains only 399 samples.
- Moderate class imbalance (60.65% bit 0 vs 39.35% bit 1).
- Limited feature set (Analog Value, Analog Voltage, NTU).
- Performance decreases as turbidity increases.

